



ENGF0034: Design and Professional Skills

Introductory lecture

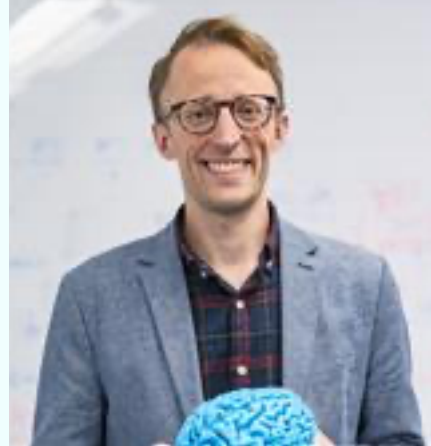
Prof. Martin Benning
6 October 2026



ENGF0034 2026/27 is a team effort



Charlotte Maughan
Jones



James Cole



Vivek Ramachandran



Amy Lourenço



Sunny Bains

Teaching Assistants



Cris Tudosie



Anya Lakhani



Shafiyyah Shafiyyah

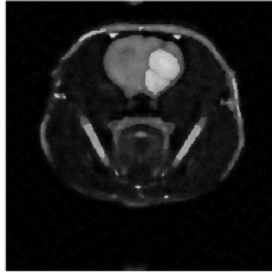
Who am I? Professor of Inverse Problems



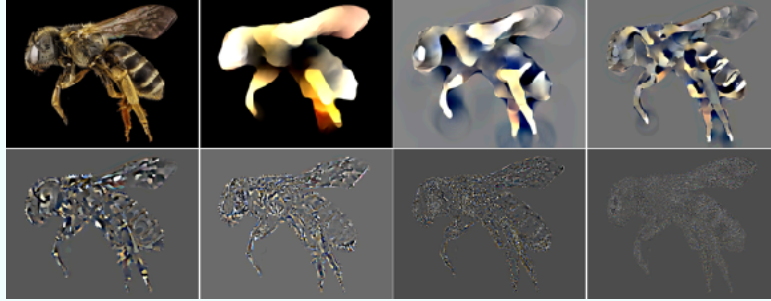
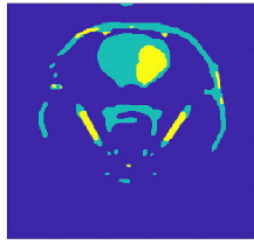
Martin Benning

Areas of expertise:

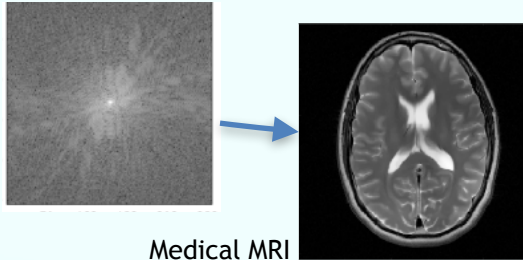
- Inverse & Ill-posed problems
- Regularisation theory
- Machine learning
- Optimisation
- Industrial & medical Imaging
- Image processing



Joint reconstruction
& segmentation



Nonlinear spectral decomposition



Medical MRI

Transmission Electron Microscopy / Tomography

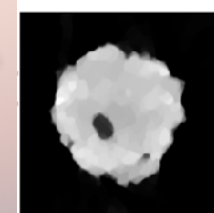
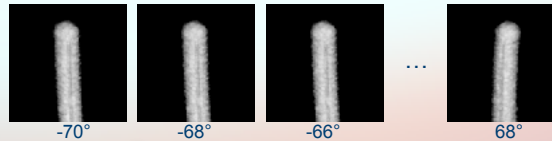
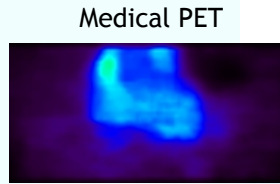
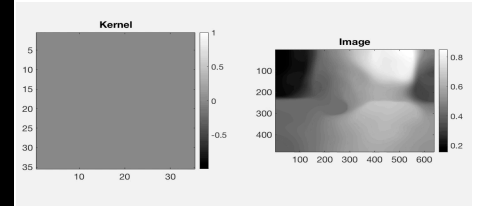


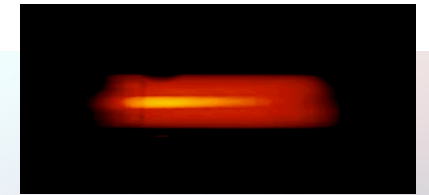
Image reconstruction



Blind image deconvolution

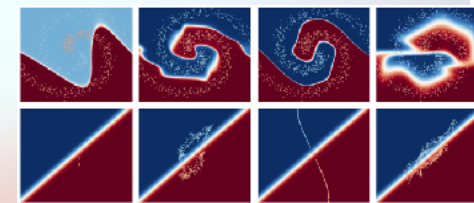


Medical PET

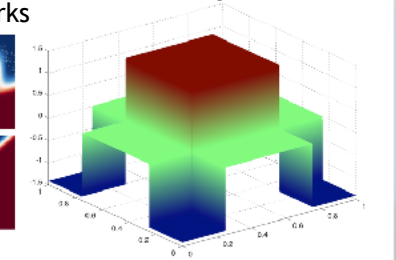


Industrial MRI

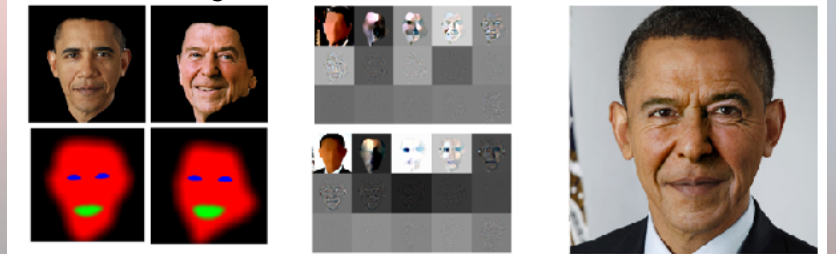
Classification with residual neural networks



Nonlinear eigenfunctions

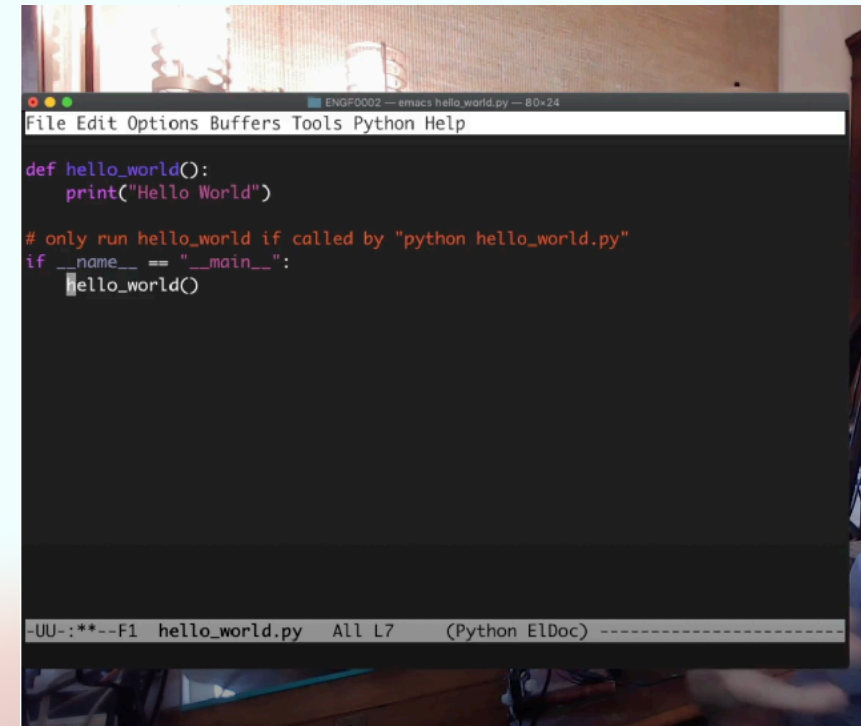


Nonlinear image fusion



Module approach

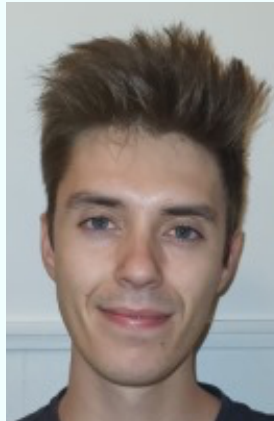
- Lectures (almost) every **Tuesday**
- Tutorials (almost) every **Friday**
- **Ask and answer questions** via the [Moodle forum](#)
- **Course materials** will be released via [git](#)
- **Assessments** will be released via git or Moodle
- **Drop-in clinics** will take place as extra support for those new to programming



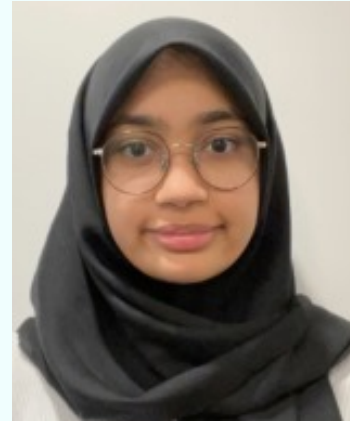
Let us checkout the module [Moodle](#) & [git](#) pages

Drop-in clinics

Teaching Assistants



Cris Tudosie



Shafiyyah Shafiyyah



Anya Lakhani

- Drop-in clinic Cris (in-person): Wednesday, 3-4 pm
G03, Henry Morley
 - Drop-in clinic Shafiyyah (in-person): Thursday, 4-5 pm
G03, Henry Morley
 - Drop-in clinic Anya (in-person): Friday 1-2 pm
G03, Henry Morley
- (Starts next week!)

Assignments

- Week 1 - 2: Introductory programming assignment
- Week 2 - 3: Debug the bomber game
- Week 3 - 5: Debug the frogger game
- Week 5 - 7: Write a tetris autoplayer (20%)
- Week 7 - 9: Write pacman protocol specification (15%)
- Week 7 in Term 2: Ethics report (15%)
- Term 2 Reading Week: Presentations (10%)
- + Scenario assessments in Term 2 (remaining 40%)

Assignments

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- Week 2 - 3: Debug the bomber game
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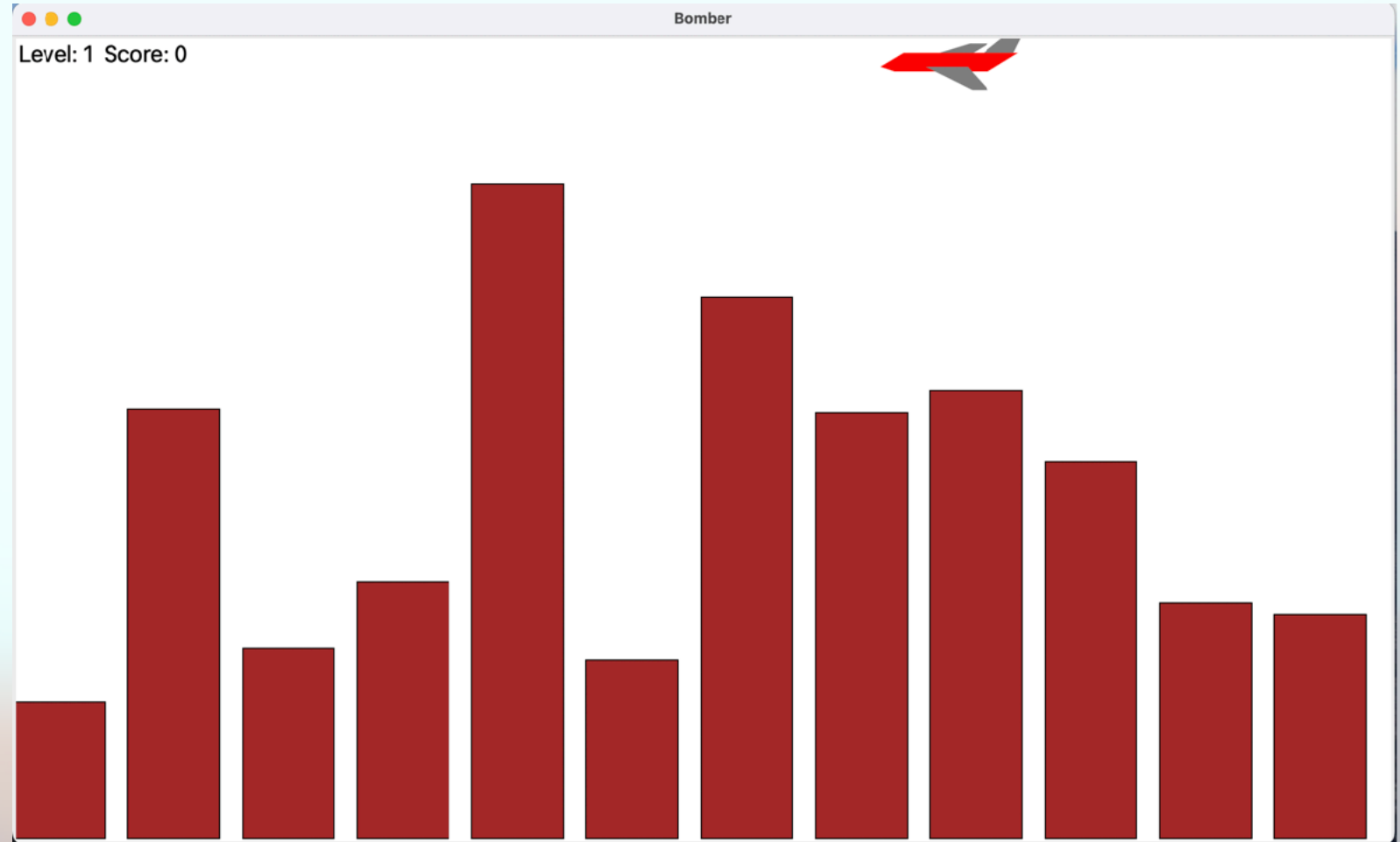
Do not plan time away
during reading week!

Assignments

Assignment 2: (Canyon) Bomber

Classic Atari Arcade Game from 1977

Task: Find five bugs that make the game unplayable, and fix them!

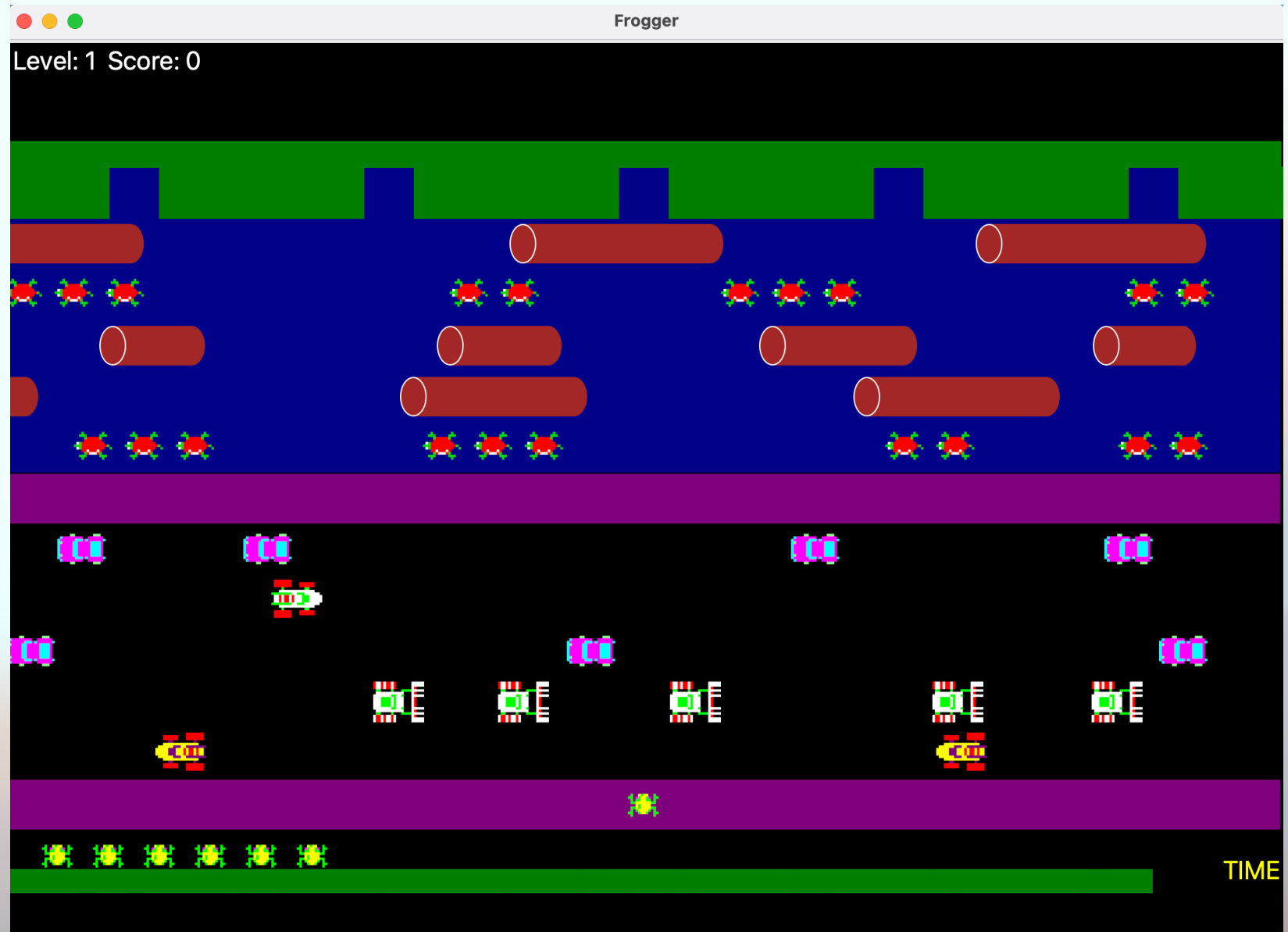


Assignments

Assignment 3: Frogger

Classic Konami/Sega Arcade
Game from 1981

Task: Find five bugs that make
the game unplayable, and fix
them!



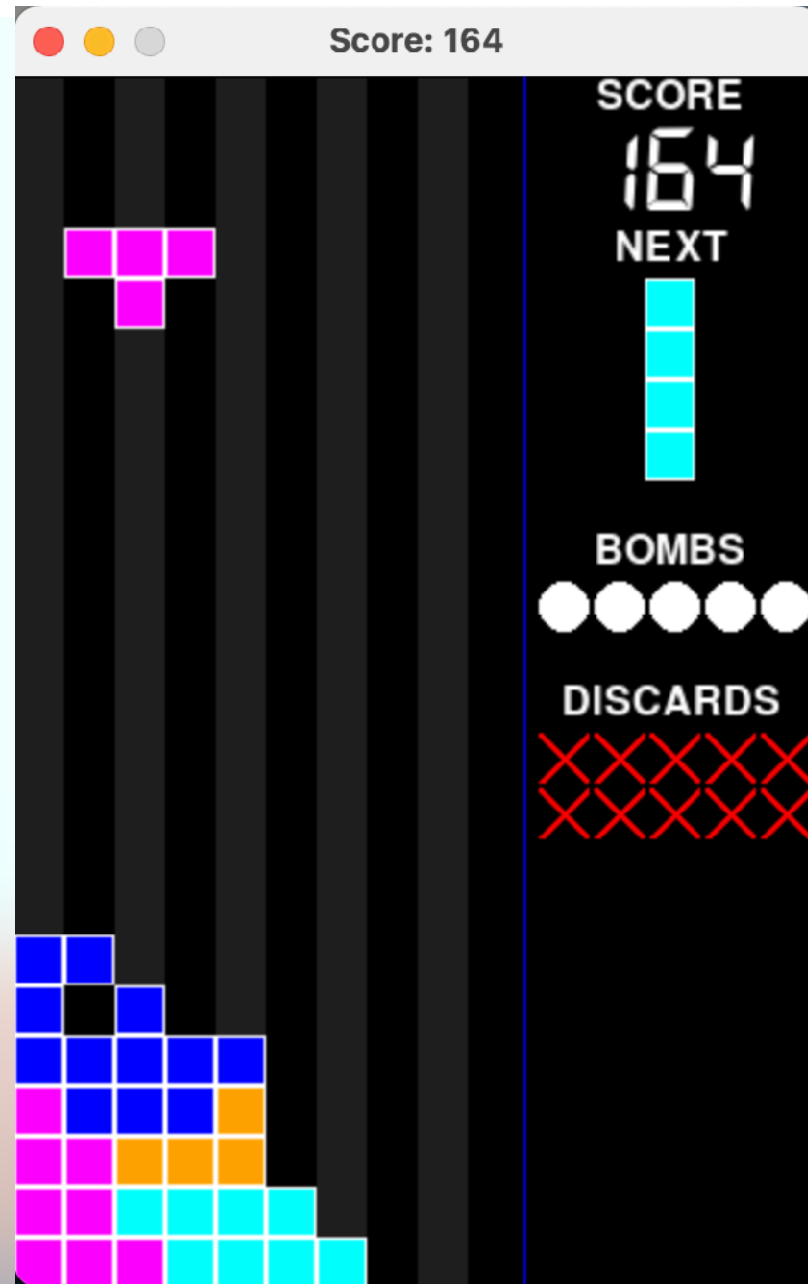
Assignments

Assignment 4: Tetris autoplayer

Puzzle video game created by
Alexey Pajitnov in 1985

Game Boy version by Nintendo
popularised Tetris in 1989

Task: write a program that
plays Tetris and compete with
your peers



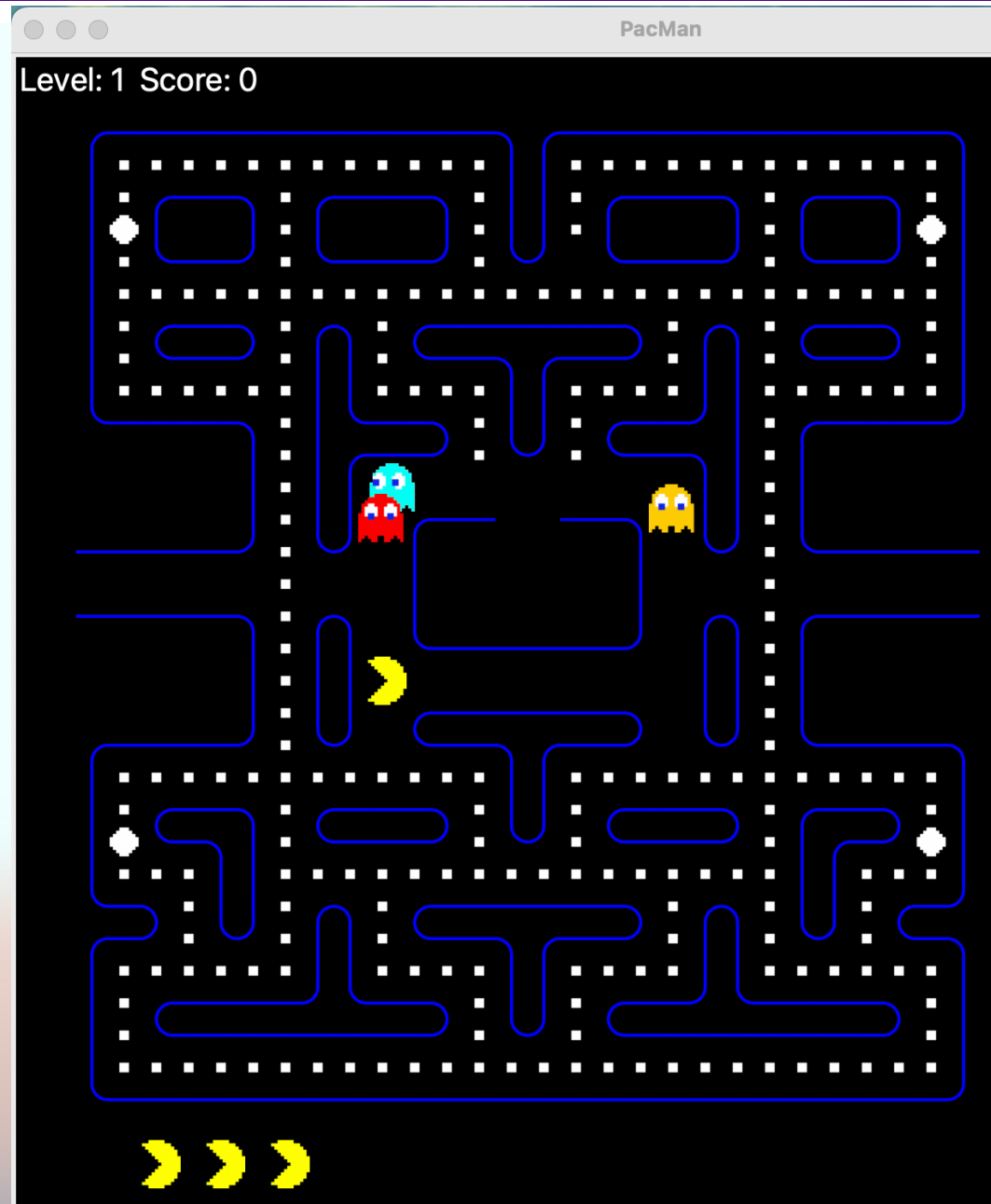
Assignments

Assignment 5: Write specification for Pacman game

1980 Maze video game

Special network version of the game

Task: write a better
network protocol
specification for this game



Other remarks

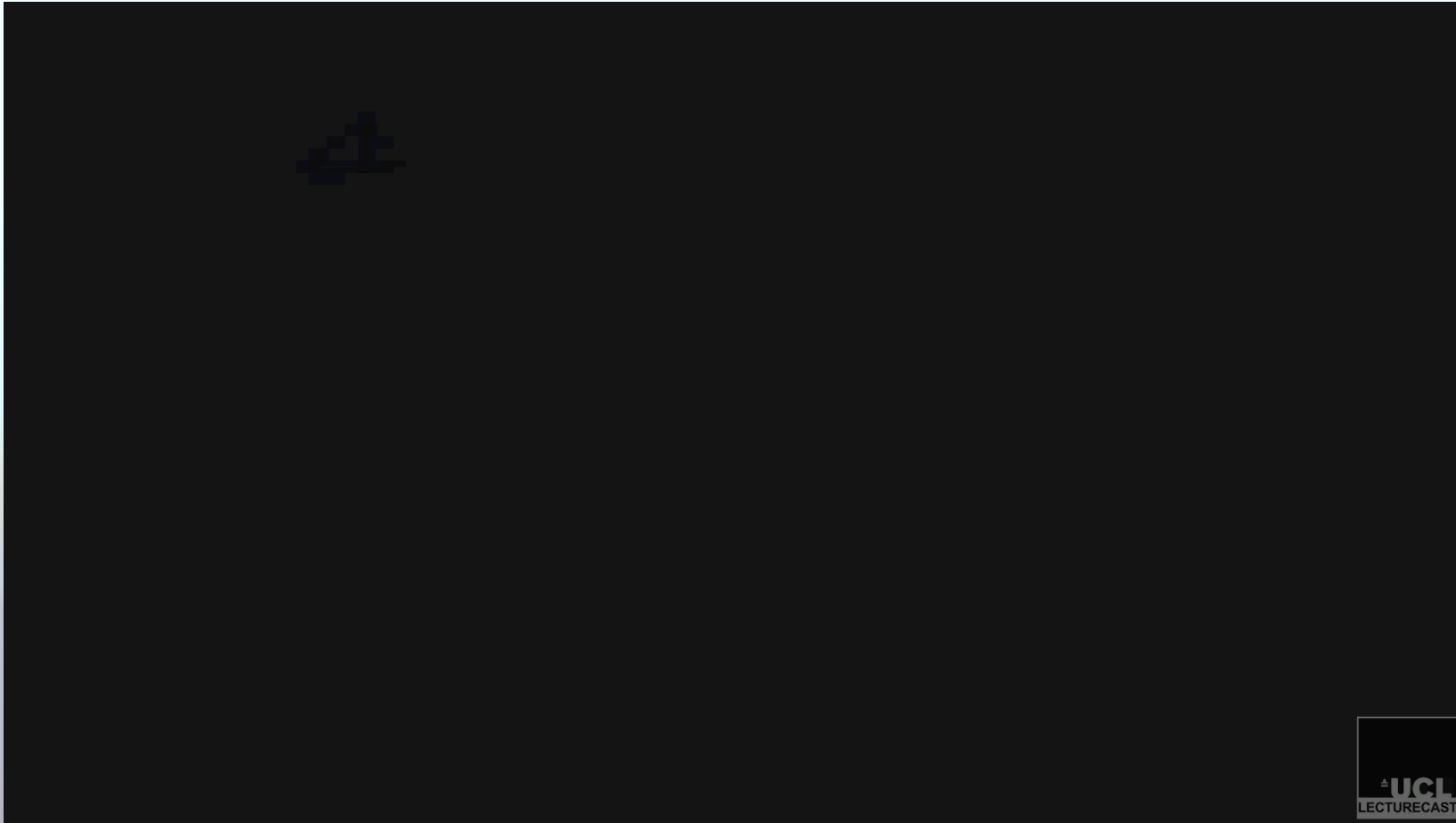
- For assessments 1 - 3 communicate and work with your peers
- If you have questions, it is likely your peers will have similar questions ➡ ask via Moodle forum, **not** via e-mail
- Do not only use forum when you have questions; help your peers and answer their questions, too!
- However, do not ask or post solutions for summative (marked) assessments (4 & 5) ➡ punishable offence
- Make use of forum and the drop-in clinics; do not suffer in silence!
- I know it is tempting to use Large Language Models (like ChatGPT, Gemini, Claude etc.), but there are risks:
 - Often you require programming experience to understand LLM outputs
 - More importantly, you require experience to identify **wrong** outputs!
 - It is not allowed to use LLMs for summative (marked) assessments!

Generative AI Use

- This module is classified as a Category 1 assessment according to the [UCL Generative AI guidelines](#).
- Category 1: GenAI tools **cannot** be used.
- You must not use LLMs, generative AI tools, code assistants, or similar systems to complete (summative assessments of) this module. You are expected to work from the course material, and standard documentation only.
- Any use of GenAI tools for summative assessments in this module is considered academic misconduct and will be handled according to UCL's standard procedures. For further details, see the [UCL GenAI assessment categories](#).

This Friday (October 9)

Pebble in the pond!



When? 11 am - 1 pm

Where?

Royal National Hotel

Next Tuesday (October 12)

Teamwork - Live Session

with Dr Charlotte
Maughan Jones



When? 11 am - 1 pm

Where?

TBA

Now / Next hour ➡ Programming foundations!